Study 2, Qwen2.5-VL-7B: The Full Battery on an A40, the Whole Corpus, and the Unquantised Arm

Engineering log, prepared for research collaboration with Professor Mustafa Abdallah

November 11, 2026

Contents

1 The Full Battery on Scholar's A40, and Stage 6 for the First Time 3

1.1 Goal . . . 3

1.2 What Was Done . . . 3

1.3 Results . . . 4

1.4 Problems and Solutions . . . 4

1.5 Standard vs. Non-Standard Choices . . . 5

1.6 Implications . . . 5

2 Every Compliant Image, Not Only the Sampled Ones 6

2.1 Goal . . . 6

2.2 What Was Done . . . 6

2.3 Results . . . 6

2.4 Problems and Solutions . . . 7

2.5 Standard vs. Non-Standard Choices . . . 7

2.6 Implications . . . 7

3 Stage 9 Completeness, Redefined on Attention (F7) 8

3.1 Goal . . . 8

3.2 What Was Done . . . 8

3.3 Results . . . 8

3.4 Standard vs. Non-Standard Choices . . . 9

3.5 Implications . . . 9

4 Stage 8 Robustness at Pre-Approved Strengths 10

4.1 Goal . . . 10

4.2 What Was Done . . . 10

4.3 Results . . . 11

4.4 Standard vs. Non-Standard Choices 11
4.5 Implications 11

5 The Unquantised Arm: Is 4-bit Compression the Cause? 12
5.1 Goal 12
5.2 What Was Done 12
5.3 Results 12
5.4 Problems and Solutions 12
5.5 Standard vs. Non-Standard Choices 14
5.6 Implications 14

6 The §5.5 Annotation Audit: Draw and Labelling Page 15
6.1 Goal 15
6.2 What Was Done 15
6.3 Results (first reader; owner review pending) 16
6.4 Scoring 16
6.5 Standard vs. Non-Standard Choices 16
6.6 Implications 17

Chapter 1

The Full Battery on Scholar's A40, and Stage 6 for the First Time

1.1 Goal

Every Study 2 number so far came from one RTX 3070 with 8 GB of memory, where a single Stage 2 pass takes about ten seconds a question. Stage 6 (stability: five sampled reruns per row) was implemented and tested but never run, because at that speed it would have occupied the card for days. This phase moves the whole battery to Purdue's Scholar cluster (NVIDIA A40, 46 GB), runs every stage there including Stage 6, and checks that the headline figures survive the change of machine.

1.2 What Was Done

Per-machine result trees. A parity run had already shown that greedy decoding reproduces exactly on one GPU but not across GPUs: on identical weights, code, data and pinned libraries the A40 agreed with the frozen 3070 baseline on 96.9% of verdicts and 42.7% of replies byte-for-byte. Every model stage therefore records the GPU in its fingerprint and refuses to resume a file written on another device, and the Scholar run writes to its own tree, `results/scholar-a40/`. Comparisons between stages (Stage 6 against Stage 2, for instance) stay inside one tree, so machine noise is never measured as instability.

Self-continuing job chains. Scholar caps a job at four hours and lets one user run two at once. Each link of a chain queues its own successor first (`afterany`), runs its stages under `timeout --signal=INT` about 25 minutes short of the wall so a stage stops between rows rather than mid-write, and on success marks itself done and cancels the successor. A real failure writes a `FAILED` marker and stops every chain for that model — there is no retry loop. Four chains (Stages 4, 5, 6 and later 8) run after the base chain (Stages 0–3, 7), and the last to finish submits the final chain (Stages 9–11).

1.3 Results

metric	RTX 3070 (frozen)	A40
sensitivity	90.9%	90.4%
specificity	40.9%	40.5%
false-alarm rate	59.1%	59.5%
rule_4 sensitivity ($n = 70$)	30.0%	25.7%
Stage 4 flip rate (black box)	6.4%	6.4%
Stage 5 attention exceeds area null	85.7%	85.4%
Stage 10 explanation correct	2.4%	2.4%

Table 1.1: The same battery on two machines. Every headline agrees within about one point and well inside its 95% interval; rule_4's four-point gap is three images out of 70.

Stage 6, run for the first time (799 rows: 400 baseline-YES, 399 baseline-NO; five reruns each at temperature 0.7):

metric	A40, 4-bit
verdict unanimous across all five reruns	51.9% [48.4, 55.4]
baseline YES / baseline NO	54.2% / 49.6%
majority of reruns overturns the greedy verdict	17.9% [15.3, 20.7]
flickering region (present in some reruns, not all)	38.0% [34.7, 41.4]
region drift > 10% of the image diagonal ($n = 612$)	36.1% [32.4, 40.0]

Table 1.2: Stage 6 stability. Stage 6's own console summary printed a median pairwise agreement of 100%, which reads as stable; unanimity shows half the rows disagree with themselves at least once.

On nearly one row in five, the verdict the paper reports (greedy decoding) is the *minority* answer under the model's own sampling distribution. The explanation is less stable still: where at least two reruns named a region, a third of the time those regions sit more than a tenth of the image diagonal apart.

1.4 Problems and Solutions

Stage 0's small-object score dropped from 89% to 78%. That is 8 of 9 images against 7 of 9 — one image. The verdict becomes "GO with limitation" on the A40, and the paper should not read anything into a one-image difference beyond that.

The "3.5× unexplained latency gap" was the local machine. On the 3070 the same images answered in 3.1 s standalone and 10–11 s inside a run, and the latency caveat in Stage 7 and the runner blamed the harness. On the A40 the in-run median is 2.3 s. The gap belongs to the 3070 machine; the caveat now quotes a standalone-to-in-run ratio only when both were measured on the same GPU.

Region drift was never computed. Stage 6 reads its rows from Stage 2, which carries no image size, so the drift column stayed empty. Stage 3's regions file has the size of every (image, rule), so the report now joins on it and recomputes drift from the stored regions; future Stage 6 runs take the sizes in-run. No other figure moved.

1.5 Standard vs. Non-Standard Choices

Standard. Bootstrap intervals on every rate; resumable, checkpointed runs.

Non-standard. Treating the GPU as part of the experimental configuration — fingerprinted, never mixed within one tree — is stricter than common practice, and it is what the 96.9% parity figure demands: 3.1% of greedy verdicts change with the card alone.

1.6 Implications

The headline results are not an artefact of one machine. Stage 6 adds a new finding for the paper: the greedy verdict is the minority answer in 17.9% of rows, and explanation regions are less stable than the verdicts they accompany.

Chapter 2

Every Compliant Image, Not Only the Sampled Ones

2.1 Goal

The Stage 1 sample is deliberate (architecture document §5.3): all 1,278 images with an annotated violation and 1,278 compliant images matched to them, 3,882 (image, rule) questions in all. That left 7,457 compliant images the model never saw. Specificity measured on the matched stratum is only as good as the matching, so this phase asks the model about every one of them.

2.2 What Was Done

`01_build_sample.py --complement` writes `complement_manifest.csv`: one row per unsampled compliant image, stratum "R", with the rule asked chosen by the same `rules_to_ask` logic as the main sample (one rule per image, as before). The manifest was built locally first (7,457 rows, SHA-256 `cccec48c...`); Scholar rebuilt it and the chain refused to start unless the hash matched, which it did byte for byte. Stage 2 then ran on it in a separate chain, writing `baseline_qwen7b-4bit_complement.csv`, so the frozen sample files are untouched.

2.3 Results

specificity	**estimate [95% CI]**	n
matched compliant stratum (C)	45.1% [42.5, 47.9]	1,278
unsampled compliant images (R)	42.6% [41.5, 43.7]	7,457
all compliant questions	43.0% [41.9, 44.0]	8,735

Table 2.1: Specificity on the whole compliant corpus (A40, 4-bit). The matched stratum is within 2.5 points of the images it was meant to stand for.

rule	**false-alarm rate, all compliant**	n
rule_1 (PPE)	57.5%	2,670
rule_2 (fall protection)	35.4%	2,713
rule_3 (unprotected edge)	93.2%	2,599
rule_4 (machinery zone)	8.5%	753

Table 2.2: False alarms by rule over every compliant question.

2.4 Problems and Solutions

None at run time. The one design choice — a single rule per unsampled image rather than all applicable rules — was kept from the main sample so the two strata are comparable; asking every applicable rule would have roughly doubled the run for a different estimand.

2.5 Standard vs. Non-Standard Choices

Standard. Validating a stratified sample against the population it represents.

Non-standard. Hash-gating the manifest across machines, so the cluster run provably asked the questions that were built and reviewed locally.

2.6 Implications

The study sample does not flatter the model: specificity is about 43% whichever compliant images are used. rule_3 is the extreme — on 93% of compliant questions the model reports a violation — and the paper should state it per rule, since the pooled false-alarm rate hides a range from 8.5% to 93.2%.

Chapter 3

Stage 9 Completeness, Redefined on Attention (F7)

3.1 Goal

Stage 9 (bounded completeness) asks whether an explanation was offered and whether it held up. As designed, "held up" meant the answer flips when Stage 4's token ablation removes the named region. Finding F4 showed that ablation fails its own negative control — a 51.7% artefact floor against a 5.9% signal — so the designed metric correctly computed a quantity with no discriminative power, and was marked not reportable (F7). This phase replaces the broken half.

3.2 What Was Done

The decision (2026-11-10) was to redefine "held up" on attention, which Stage 5 already measures without any intervention and which survived its own null (F6). `09_completeness.py --basis attention` classifies each row:

- **no usable explanation** — no region was named;
- **supported** — the share of attention mass inside the named region exceeds the share of the image the region covers (lift > 1);
- **weak** — a region was named but attention inside it is at or below its area share.

It writes `completeness_attention_{model}.csv`. The ablation basis stays the default so the frozen file reproduces, and its numbers are never reported. The report heads the section "attentional, not causal" and prints the share of regions at a lift of 2× or more beside the headline.

3.3 Results

class	RTX 3070	A40
explanation supported	67.0%	66.5%
explanation weak	11.1%	11.4%
no usable explanation	21.9%	22.2%
lift $\geq 2\times$ (regions only)	43.4%	42.6%

Table 3.1: Stage 9 on the attention basis, $n = 3{,}882$ (regions only: 3,031 / 3,022). Median lift 1.74 / 1.73.

3.4 Standard vs. Non-Standard Choices

Non-standard, and it must be argued in the paper. The claim changes from causal ("removing the region changes the answer") to attentional ("the model looked where it said"). That is a weaker claim, and the paper states it as one. "Supported" is crossed by any lift above 1.0, which is why the 2× line is reported beside it: under half the named regions draw twice their area's share of attention.

3.5 Implications

Completeness is reportable again, on a basis that has passed a null test. A metric defined over another metric inherits that metric's validity; this one now inherits Stage 5's, which was checked, rather than Stage 4's, which failed.

Chapter 4

Stage 8 Robustness at Pre-Approved Strengths

4.1 Goal

Does a verdict, and the region that explains it, survive the kind of image degradation a site camera produces? Stage 8 had not been built. This phase builds it, with the perturbation strengths fixed and approved before any run, so they could not be tuned against results.

4.2 What Was Done

Five perturbations, applied one at a time, never combined:

perturbation	**strength (approved 2026-11-10)**
blur	Gaussian, $\sigma = 1\%$ of the image's long side
low light	-2 EV, applied in linear light, not gamma-encoded values
occlusion	a mean-luminance grey block over the centre 50% $\times$ 50% of the image
contrast	factor 0.4
desaturate	ITU-R 601-2 luma

The subset is 1,600 Stage 2 rows (800 YES, 800 NO, at least 40 per rule), seed 42; each row is asked again under each perturbation with greedy decoding. Per perturbation the stage records the new verdict, whether it matches the baseline, the region, and the baseline-to-perturbed centroid distance over the image diagonal. A surviving verdict whose region moved more than 10% of the diagonal is flagged — the same cut-off Stage 6 now uses. For occlusion it also records how much of the annotated hazard the block hides, so the case where the block covers the hazard itself is reported separately. The strengths are part of the run fingerprint.

4.3 Results

perturbation	verdict survives	survived, region moved $> 10\%$
blur	70.4% [68.2, 72.6]	47.7% ($n = 459$)
low light	90.7% [89.2, 92.1]	13.8% ($n = 884$)
occlusion	80.2% [78.2, 82.1]	35.1% ($n = 643$)
contrast	90.4% [88.9, 91.8]	14.8% ($n = 865$)
desaturate	86.6% [84.9, 88.2]	20.9% ($n = 789$)

Table 4.1: Stage 8, A40, 4-bit, $n = 1{,}600$ per perturbation. The drift column is over rows that survived and named a region both times.

Occlusion that hides at least 10% of the annotated hazard leaves the verdict standing 67.9% of the time ($n = 290$); occlusion that hides less leaves it standing 85.1% ($n = 141$). That is the direction expected if the model's verdict depends on the hazard rather than on the scene around it.

4.4 Standard vs. Non-Standard Choices

Standard. One-factor-at-a-time perturbation; survival rate per perturbation, never pooled.

Non-standard. Strengths fixed and approved before any run; low light simulated in linear light; the occluder's coverage of the annotated hazard recorded per row, so occlusion results separate "hid the evidence" from "hid something else".

4.5 Implications

Mild blur is the most damaging change: three verdicts in ten flip, and when the verdict survives, the region moves in almost half the cases. A surviving verdict is not a surviving explanation — the paper should report both columns.

Chapter 5

The Unquantised Arm: Is 4-bit Compression the Cause?

5.1 Goal

Every Study 2 result used Qwen2.5-VL-7B compressed to 4 bits (nf4) to fit the 8 GB card. A reviewer can ask whether the poor explanation results are a property of the model or of the compression. The A40 has room for the unquantised model in bfloat16, so this phase runs the full battery on it.

5.2 What Was Done

A new tier, `qwen7b-bf16`, loads the same checkpoint folder at the same pinned revision (`cc594898...`) with quantisation off. The chains take the model as a parameter; a non-primary tier gets its own state directory and model-suffixed output files in the same A40 tree, so the two tiers compare without machine noise. Every stage ran: 0–10, plus 8 and the attention basis of 9.

5.3 Results

Table 5.1 sets the two tiers side by side.

Compression changes *what* the model answers — most of all on rule_4 (a worker in the danger zone of heavy machinery), where the unquantised model finds three times as many annotated violations, with intervals that do not overlap. It barely changes *how good the explanations are*: every explanation metric moves by two points or less. The unquantised model is also less steady under sampling, especially on its NO verdicts.

5.4 Problems and Solutions

Stage 6 stopped on its parse-rate guard. After 100 rows only 90% had all five sampled replies readable, below the pre-registered 95% floor. The first hypothesis — the 128-token reply cap cutting the reply off before the final ANSWER line — was wrong: a 200-call diagnostic found the longest reply was 94 tokens. Instead, at temperature 0.7 the model sometimes replies with the single word "NONE" (11 of 13 failures) or "ANSWER: MAYBE". Per reply that is 2.6%; but one bad reply in five failed the whole row.

Decision (2026-11-10): grade the guard per reply, as Stage 2's is. The guard exists to catch a degrading prompt, which a per-reply rate still does. An unreadable reply still counts as a disagreement in the agreement rate, so no 4-bit figure changed. The report gives the unreadable-reply rate on its own line. bf16 Stage 6 was rerun from scratch; the tripped attempt is kept in

metric	**4-bit**	**bf16**
what the model answers		
sensitivity	90.4%	93.8%
specificity	40.5%	36.0%
rule_4 sensitivity ($n = 70$)	25.7% [15.7, 35.7]	81.4% [71.4, 90.0]
Stage 0 small-object ($n = 9$)	78%	89%
how good the explanation is		
Stage 10 explanation correct	2.4%	2.1%
Stage 5 attention exceeds area null	85.4%	85.9%
Stage 9 explanation supported	66.5%	68.3%
Stage 4 flip rate (black box)	6.4%	6.8%
Stage 8 survives blur / occlusion	70.4 / 80.2%	68.8 / 79.4%
stability under sampling (Stage 6)		
verdict unanimous	51.9%	47.0%
baseline NO	49.6%	36.8%
unreadable sampled reply	0.7%	2.3%
in-run median latency	2.34 s	2.02 s

Table 5.1: The same checkpoint, A40, same questions, with and without 4-bit compression.

`results/scholar-a40/superseded/`. This is a change to a pre-registered guard after it tripped, and the paper states it as such.

One Stage 8 row short. bf16 has 1,599 rows, not 1,600: the shared seeded sampler rounds each rule's proportional share, and for bf16's NO verdicts the shares round to 799. The sampler is shared with Stages 4 and 6 and their files are frozen, so it is left as is and noted. One further blur reply out of 7,995 was unreadable.

5.5 Standard vs. Non-Standard Choices

Standard. An unquantised ablation of a quantised model.

Non-standard. Running it through the identical battery on the same GPU, so the comparison isolates compression from machine, data and code.

5.6 Implications

The poor explanation quality reported for Study 2 is not a compression artefact: the unquantised model explains no better. The paper can say so with a measured comparison. Accuracy is a different matter — rule_4's sensitivity under 4-bit is a compression effect and should be reported as one.

Chapter 6

The §5.5 Annotation Audit: Draw and Labelling Page

6.1 Goal

The dataset grades its own annotations, and 47% of images are self-flagged "poor info". Every Study 2 figure is measured against those annotations, so they are a reference standard, not ground truth. Architecture document §5.5 asks a person to relabel 100 images, 50 per quality tier, so that the annotation error rate can be reported beside every result.

6.2 What Was Done

The draw. `12_audit_sample.py` draws from the whole corpus (10,013 images), not the study sample. The pre-registered draw was a pure random 50 per tier with seed 42. It held 3 hazard images in 100, too few to estimate how often a *marked* violation is wrong. **Decided 2026-11-10:** stratify by hazard within tier — 25 images with an annotated violation and 25 without, per tier, each pool drawn with its own seeded generator and independent of record order. Corpus-level error rates are reweighted to the true 12.8% hazard prevalence. The 50 hazard images carry 38 rule_1, 7 rule_3, 3 rule_2 and 3 rule_4 violations, so per-rule rates beyond PPE will rest on very few images. 61 of the 100 images were in the study sample and carry a model answer.

The labelling page. A private web page presents each image, downscaled to a 1,024-pixel long side. Step one is blind: the auditor answers the four rule questions (yes / no / can't tell) from the photograph alone and saves. Only then does the page reveal the dataset's annotation boxes and the model's answer and region, and the blind answers lock. Step two asks, per annotated box, whether it sits on its hazard (yes / partly / no), with a free-text note. Each auditor's answers are stored under their own identity; the owner can read all of them.

Decided 2026-11-11: a frontier model is the first reader. No human auditor was available, and the owner chose to have Claude (Opus 5.5) answer the audit, with the owner reviewing every answer afterwards. The procedure kept the page's blind-then-reveal order. The 100 images were put in a seeded random order (seed 2026) so that the hazard and compliant strata were not grouped. Each image was viewed at the same 1,024-pixel long side and answered for its applicable rules against the dataset's own rule text, with a one-sentence reason, before any annotation was opened. The blind answers were committed to the repository (`7b4adeb`) before the reveal. Only then were the annotation boxes drawn on the 50 hazard images and each judged for placement. Answers are stored in the format the labelling page writes (`results/audit_answers/audit/claude-opus-5-5/`), so the scoring script runs on them unchanged. A second page, `audit/review.tpl.html`, shows each answer beside the annotation, disagreements first, and records the owner's final answer per rule.

measure	rate [95% CI]	n
marked, not confirmed	10.0% [2.0, 18.0]	50
unmarked, auditor sees one	8.3% [5.3, 11.7]	265
can't tell	6.5% [3.9, 9.2]	337
image disagrees, reweighted to the corpus	16.9% [8.9, 25.9]	100
annotated box on its hazard	47 yes, 4 partly, 0 no	51

Table 6.1: Section 5.5 audit, first reader Claude Opus 5.5, question level unless stated.

6.3 Results (first reader; owner review pending)

Table 6.1 gives the scores. They are disagreements between one reader and the annotation, not yet established annotation errors.

Marked, not confirmed (5 of 50). Each was looked at after the reveal. One is the reader's own miss: two workers in shorts (image 0001305), a rule_1 violation under the dataset's rule text that the reader overlooked. One is a filing difference: the reader called two workers climbing a steep slope a fall-protection case, and the annotation called it an unguarded edge (0000991). One is a distant figure the reader saw as helmeted (0002260). Two are rule_3 judgements where the reader saw a graded slope or a fence along the rim (0004032, 0007712).

Unmarked, auditor sees one (22 of 265). Half are rule_4. Eleven times the reader saw a worker within a few metres of an operating excavator that the annotation left unmarked. Nine of those were on images annotated only for another rule, usually a missing hard hat on the same worker. Next come rule_3 (6), rule_2 (3) and rule_1 (2). The annotation looks reliable for PPE and incomplete for plant proximity.

Box placement. No annotated box was off its hazard. Four were partly off, all rule_1 boxes drawn around a group that included helmeted workers.

6.4 Scoring

The scoring script, `13_audit_score.py`, was written and tested ahead of the answers. It keeps two errors apart — *marked, not confirmed* (the annotation marks a violation the auditor does not see) and *unmarked, auditor sees one* — reports "can't tell" on its own line, and scores only rules that apply to the image. Its corpus figure reweights the 50/50 draw to each tier's real hazard prevalence (rich info 771 of 5,271 images, poor info 507 of 4,742) and the tiers to their real shares, with a bootstrap that resamples images within each (tier, stratum) cell. Per tier, "marked, not confirmed" rests on about 25 images and falls below the $n = 30$ reporting floor by design, so it is quoted pooled across tiers.

6.5 Standard vs. Non-Standard Choices

Standard. Blind relabelling of a reference standard; stratified sampling with reweighting.

Non-standard, and stated in the paper. The stratified draw replaces the pre-registered pure random draw, decided after seeing that the random draw could not answer the question it was for. Blind-then-reveal in one page keeps the first answers uncontaminated while still collecting box-level

judgements.

Non-standard, and stated in the paper. The first reader is a vision-language model, not a person. It is a different and larger model family than the one under study, but it can share that kind of model's blind spots, chiefly distant, few-pixel workers (10 of its 22 "can't tell" answers were on such figures). The owner's review of every answer is what makes the audit a human one. The paper reports the first reader's figures, the owner's adjudicated figures, and how often the two differ.

6.6 Implications

Every accuracy and explanation figure in Study 2 is "agreement with the dataset's annotations". The first reader puts the corpus-level image disagreement at about one in six, concentrated in rule_4. Two Study 2 readings depend on that. Some of the model's rule_4 "false alarms" on compliant images may be real hazards the annotation left out. The unquantised model's higher rule_4 sensitivity is measured against an incomplete rule_4 reference. Neither changes a frozen number. Both are stated beside the rule_4 results, and the owner's review fixes the size of the effect.